CYCLISTIC BIKE-SHARE

Annual Membership Conversion Analysis

Analysis Period: 12 months | Total Rides Analyzed: 5,601,662

Business Task

- Analyze how annual members and casual riders use Cyclistic bikes differently.
- Use data-driven insights to recommend marketing strategies that convert casual riders into members
- Present findings with visualizations to earn executive team approval.

Data Sources

The analysis uses Cyclistic's historical trip data covering the most recent 12 months of rides. The dataset is sourced from Motivate International Inc.

Data Characteristics

Attribute	Description
Time Period	Last 12 months(Mar'25 – Feb'25)
Total Rides	56,01,662 recorded trips
Fields	Ride ID, Bike type, Member type, Start/End Time
Format	CSV files(one per month), merged via Excel Power Query
Privacy	No personal identifiable information

Data Credibility

- Reliable: Collected directly from Cyclistic's tracking system across 692 stations
- Original: Primary source data, no third-party aggregation
- Comprehensive: Covers all 12 past months with over 5.6 million records
- Current: Reflects the most recent 12 month-period
- Cited: Publicly licensed from Motivate International Inc.

Limitations

- Cannot connect pass purchases to individual riders
- Cannot determine whether the casual riders live in the service area
- Cannot identify repeat casual-riders vs first-time users

DATA CLEANING

Tools Used

Microsoft Excel: Power query for merging, pivot tables for aggregation and pivot charts for visualizations.

Cleaning Steps

1. Downloaded 12 individual monthly CSV files and merged them into a single dataset using Power Query.
2. Removed unnecessary columns not relevant to the analysis.
3. Created derived columns: Month name, Day name, Hour, Ride time((ended_at-started_at) * 60 * 24).
4. Created a custom column(Ride time categories) from the ride time column with four brackets: Under 10 mins, Under 30 mins, Under 60 mins and Over 60 mins.
5. Replaced NULL results in Ride Time column with “Unknown”.

Data Integrity Verification

1. Confirmed no duplicate Ride IDs.
2. Verified start times precede end times for all records.
3. Verified that the members_casual column contains only “casual” & “member” values.
4. Total 5,601,662 count of records across all the pivot aggregations.

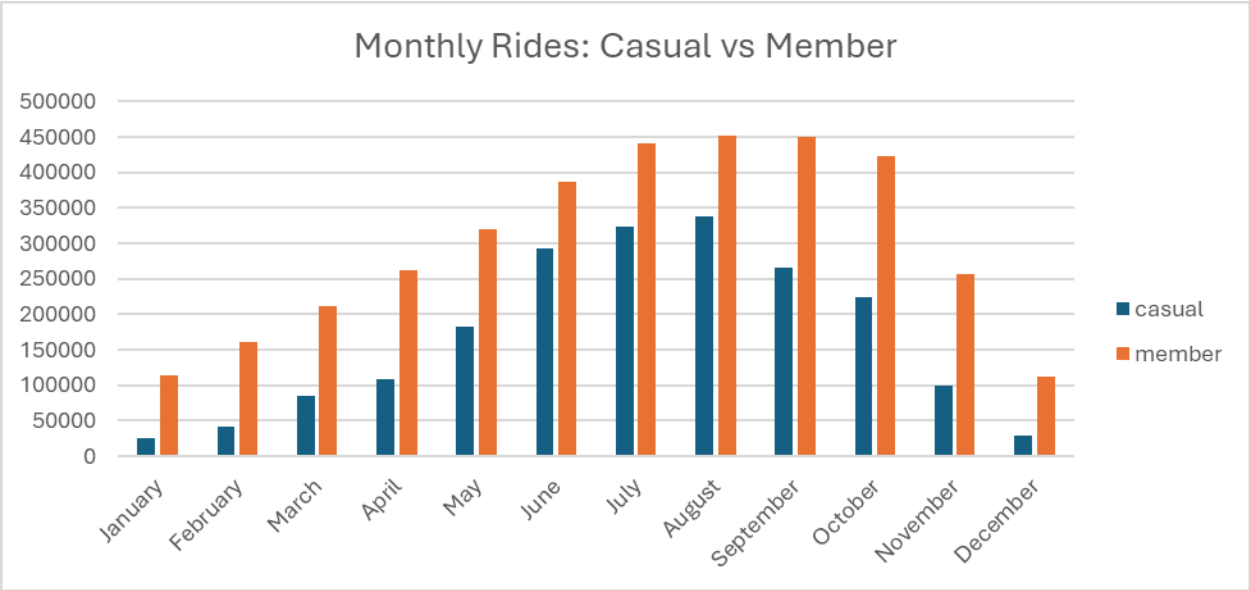
Findings

Overall Ride Volume

Rider Type	Total Rides	Share of Total
Casual	2,013,512	35.9%
Member	3,588,150	64.1%
Total	5,601,662	100%

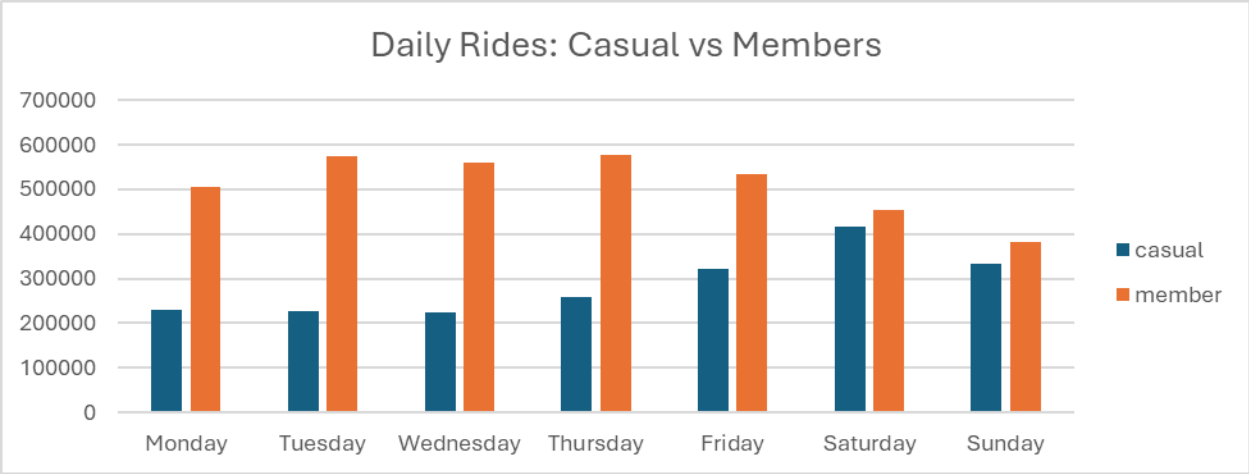
Annual members account for nearly two thirds of all rides. With over 2 million casual rides per year, there is a substantial opportunity for conversion.

Finding 1 – Yearly usage(Month wise)



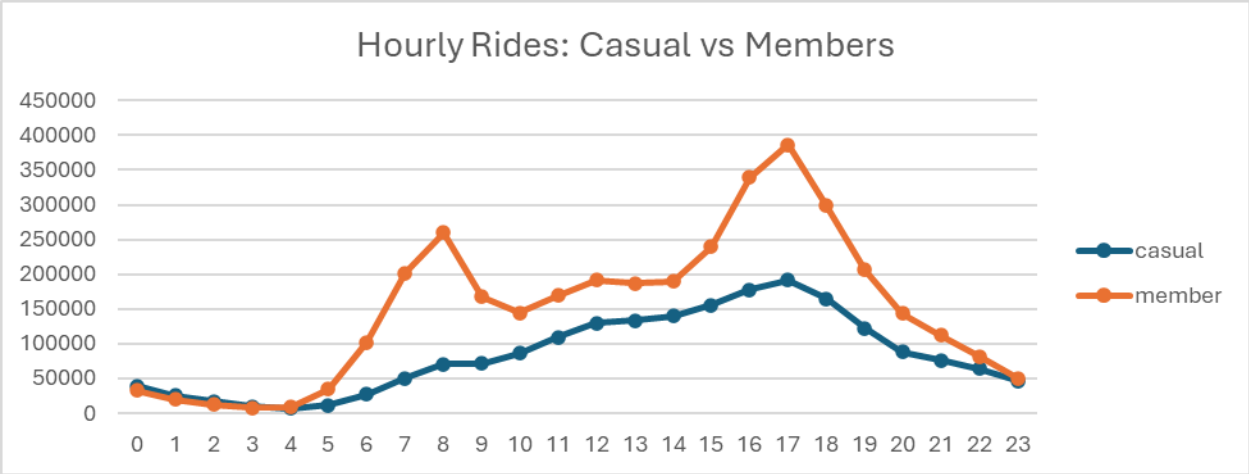
- Both Casual members' and Annual members' ride usage peak in summer(May – October).
- Nearly 81% of total casual rides and 70% of annual members' rides occur in May – October.
- Annual members ride more consistently throughout the year, indicating their reasons to choose to ride(commuting to and from work).

Finding 2 – Weekly Usage(Day wise)



- Members ride more on the weekdays**(Mon-Friday)**, suggesting that they use the rides to commute to work.
- Casual members ride the most on the weekends**(Fri-Sun)**, suggesting that they use the rides for recreational purposes.

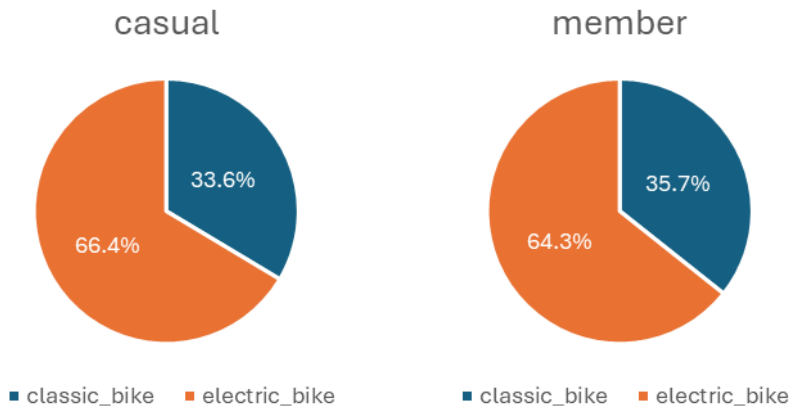
Finding 3 – Daily Usage(Hour wise)



- Member rides peak during the working commute hours(7-9AM & 3-7PM). This doubles down on the finding 2, that suggested that members ride the most to and from work.

- Casual rides have a gradual peak at around 5PM, solidifying the finding 2 that suggests that casual riders ride the most for recreational purposes.

Finding 4 – Bike Type Usage

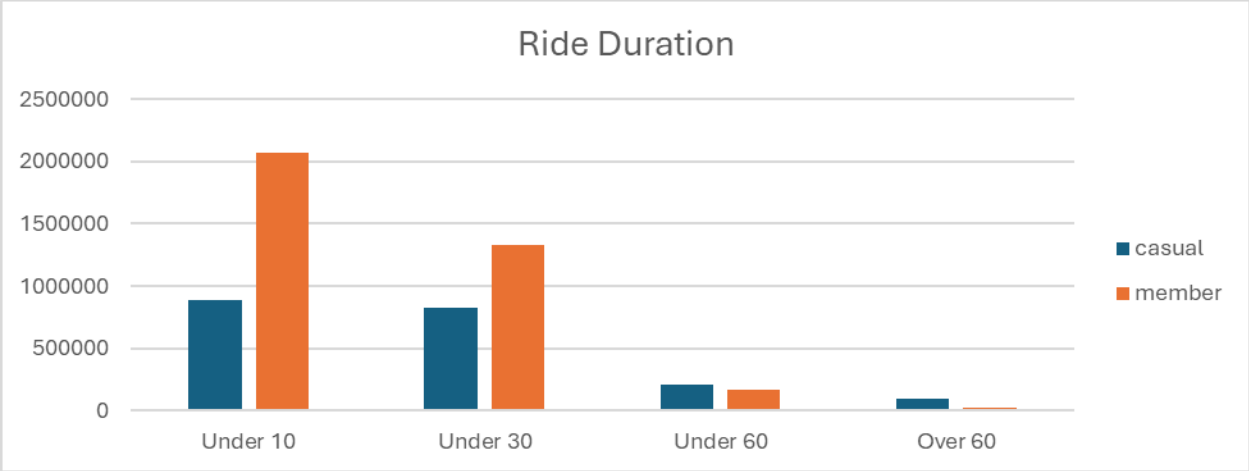


Bike Type Preference

Bike Type	Casual Rides	Casual %	Member Rides	Member %
Classic	676,160	33.6%	1,280,370	35.7%
Electric	1,337,352	66.4%	2,307,780	64.3%

- Electric bikes are preferred over classic bikes in both casual and annual members which is a result of the comfort and reduced fatigue in electric bikes.

Finding 5 – Ride Duration



- Members strongly favor short rides(57.7% under 10 mins) reinforcing the hypothesis that they majorly commute to and back from work.
- Casual riders have a broader distribution, with majority of the rides under 30 mins but nearly 5 times more likely to ride more than 60 mins than annual members.

Bike Type	Casual Rides	Casual %	Member Rides	Member %
Under 10	888,074	44.1%	2,070,829	57.7%
Under 30	820,596	40.8%	1,324,432	36.9%
Under 60	207,406	10.3%	169,257	4.7%
Over 60	97,436	4.8%	23,632	0.7%

Key Findings Summary

Key Insights at a Glance
• Members commute: sharp peaks in usage at 8AM & 5PM on weekdays.
• Casual riders recreate: weekend heavy, gradual evening peaks & longer trips.
• Summer is peak season for both types: May – Oct
• Electric bikes are preffered over classic bikes by both groups.
• Members take short, fast trips. Casual riders take long exploratory rides.

Behavioral Profiles-

Dimension	Casual Rider	Member Rider
Peak season	Summer(Jun – Aug)	Summer(May – Oct)
Peak Days	Fri – Sun	Mon – Fri
Peak hour	4 – 6 PM(gradual)	8 AM & 5 PM(sharp spikes)
Preffered Bike Type	Electric	Electric
Ride Durations	10 – 30 mins	Under 10 mins
Usage type	Recreational	Commute

Top 3 Recommendations

The following recommendations are designed to convert casual riders to annual members:

Recommendation 1 – Weekend & Seasonal Passes

Introduce a "Weekend Pass" and a "Summer Pass" as membership tiers priced between the ride/day and annual memberships. These lower commitment products can include an upgrade incentive(credits for extra usage if converted within 90 days).

Implementation
• Launch "Weekend pass" before the spring season(Mar – Apr) to capture early summer riders.
• Offer a "Summer Pass"(May – Oct) with an upgrade to annual prompt in the app.
• Use in-app notifications and emails to nudge Weekend/Summer pass holders with a conversion discount in September before they churn.
• Target ads on Fridays and weekends when casual riders are the most active.

Success Metric: **>=15%** of Weekend Pass holders upgrade to annual membership within 90 days of purchase.

Recommendation 2 – "Try Commuting" Campaigns

Annual members commute to and from work using Cyclistic bikes. Casual riders don't exhibit such patterns. A targetted campaign that educates casual riders on the benefits of annual memberships and how it pays for itself within weeks can increase conversion.

Implementation
• Social media ads highlighting the savings and how the annual membership pays for itself within weeks.
• Promote free-trial weekday rides for casual riders during off-peak periods.
• Partner with employers and gated societies to promote usage.

Recommendation 3 – Long-Ride Incentives

Introduce a loyalty program for longer rides(e.g., points for rides over 30 mins) and discounts based on ride frequency.

Implementation
• Ride Rewards Program: Incentivize longer rides(over 30 mins) with credits redeemable as extra ride time or merchandise.
• Provide discounts for "casual to member" upgrades specifically for riders who have completed 5+ long rides(over 30 mins) in a month.

Recommendation 4 - Electric Bike Promotions

Since, both groups prefer electric bikes, featuring electric bikes as a premium membership benefit, adds value.

Implementation
• Highlight electric bike availibility as "Always available" with memberships and "limited availibility" for casual riders.

Recommendation 5 – Evening ad campaigns

Since casual riders ride mostly for recreational purposes in the evenings, a targetted ad campaign at 4-6pm in the weekdays can build habit.

Next Steps & Additional Data

Recommended next steps

- Partner with the product team to build a cost-comparison calculator.
- Design A/B tests for the Weekend and Summer pass pricing tiers..
- Develop digital media campaigns targetting casual riders on weekdays and Friday afternoon.

Additional Data that would strengthen the Analysis:

- Geographic data: knowing where most rides occur and where the gap is, gives a better idea of the consumers and where to target the physical ad campaigns.
- Demographic data: Age, gender and zipcode to refine audience targetting.
- Survey data: Direct feedback from the consumers helps understand their pain points and success metrics,i.e, why one would or wouldn't subscribe to a membership.
- Pricing sensitivity data: knowing how much people are willing to pay for memberships can help setup the correct pricing tiers.